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Accelerating Digital Transformation with Automated Pressure Transient Analysis

M. Y. Martinez, A. Al Shalabi, and B. A. K. Alkindi, ADNOC Onshore, Abu Dhabi, United Arab Emirates; U. Tariq, 2KAPPA engineering, Abu Dhabi, United Arab Emirates; A. Kumar, ADNOC Onshore, Abu Dhabi, United Arab Emirates; S. Duchenne, Total Energies, Abu Dhabi, United Arab Emirates; M. Hasnat, K. R. S. R. Al Naqbi, A. Diwakar, and S. Anbalagan, ADNOC Onshore, Abu Dhabi, United Arab Emirates; V. Vaucherey, 2KAPPA engineering, Abu Dhabi, United Arab Emirates

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Abstract

This paper outlines the outcomes of a pilot project focused on automating Pressure Transient Analysis (PTA) by utilizing real-time data from Permanent Downhole Gauges (PDGs). The main objective was to enhance the efficiency of well performance evaluation, estimate static pressure, and streamline the identification of potential interventions through a cloud-based digital platform. The scope covered the integration of real-time data acquisition, event detection, automated PTA, and data processing, highlighting the role of digital transformation in improving engineering workflows and decision-making.

The pilot utilized a cloud-based platform that automatically gathered real-time pressure data from PDGs installed in active wells. Smart filtration techniques were applied to reduce noise while preserving data signature. The system identified key events, such as shut-ins, synchronized pressure and production data, and initiated automated Pressure Transient Analysis (PTA) when the necessary data was available. Results were validated against traditional methods and formatted for integration with other systems.

Key challenges included ensuring compatibility with existing infrastructure, training engineers to adopt new digital tools, and addressing cybersecurity concerns associated with cloud operations. Collaborative work between the multidisciplinary teams from the operator and service provider was essential for refining workflows, incorporating enhancements, and customizing the solution to meet specific operational requirements. This cooperation maximized the solution's effectiveness, ensuring successful implementation.

The automation of PTA yielded notable benefits, including more than 50% reduction in manual analysis time. The real-time integration of pressure and production data facilitated ongoing well monitoring and enabled faster identification of potential intervention needs. Smart data filtration reduced overall data volume by 70%, while preserving essential information for accurate analysis. Automated PTA provided quick and reliable insights into well performance, such as estimates of static pressure and flow behavior, which were validated using traditional techniques.

The implementation of this digital solution underscored the importance of integrating automation into routine engineering tasks. Engineers gained valuable time to focus on more complex activities, while the system efficiently handled data processing and analysis. The project revealed some initial challenges, such as the need for workforce training and the adoption of secure data protocols for cloud environments. However, these were addressed through targeted training programs and enhanced data security measures.

This pilot demonstrated that automating PTA, supported by digital transformation, not only optimizes well performance evaluation but also drives improvements in the overall decision-making process, helping operators to make data-driven decisions faster.

Introduction

The oil and gas industry is undergoing a profound digital transformation, driven by the need to enhance operational efficiency, reduce manual workloads, and accelerate data-driven decision-making. Among the many disciplines impacted by this shift, reservoir engineering stands out due to its reliance on high-frequency data and complex analytical workflows. Pressure Transient Analysis (PTA) remains a cornerstone of well performance evaluation, offering critical insights into reservoir behavior, static pressure, and flow regimes.

Permanent Downhole Gauges (PDHGs) have become essential tools for continuous reservoir monitoring, enabling engineers to capture bottomhole pressure data in real time. However, the traditional approach to PTA, often manual and time-consuming, limits the frequency and responsiveness of analysis, especially across large well inventories.

To address these limitations, a pilot project was initiated to automate PTA using real-time PDHG data. A strategic partnership was formed with a technology vendor to implement the solution, ensuring alignment with operational requirements and integration with existing infrastructure.

A pilot project was initiated to automate PTA using data from 30 commissioned PDHGs and 50 installed gauges across 30 wells, 29 oil producers and 1 water injector ([Martinez Santiago, et al. 2025](#)). It is aimed to streamline well performance evaluation, estimate static pressure more efficiently, and enable faster identification of potential interventions. This paper presents the methodology and results of the pilot implementation, demonstrating how automated Pressure Transient Analysis (PTA) can be applied to enhance well management- reservoir practices.

Methodology

Digital Transformation Operations in Oil and Gas

The pilot project was launched as a strategic initiative within a broader artificial intelligence (AI) transformation framework aimed at modernizing reservoir management practices. Recognizing the inefficiencies of manual workflows in pressure transient analysis (PTA), the company developed a comprehensive AI strategy focused on automation, data integration, and cross-functional collaboration. This strategy was designed to streamline engineering tasks, reduce interpretation delays, and improve the fidelity of reservoir diagnostics.

A cloud-native automation platform was selected to orchestrate PTA workflows across multi-well assets. This platform enabled real-time data ingestion, event detection, and model execution, while supporting customizable extensions through scripting interfaces. Its modular architecture allowed engineers to tailor workflows to specific reservoir conditions, facilitating scalable future deployment and seamless integration with existing asset management systems.

Effective communication between internal stakeholders including reservoir and IT engineers, field operations, and the external technology vendor was critical to the success of the pilot. Together, these teams defined operational requirements, validated analytical outputs, and ensured cybersecurity compliance.

This collaborative approach fostered transparency and accelerated the adoption of digital tools across the organization.

The pilot demonstrated how automation could transform internal processes in reservoir development. By reducing manual analysis time and enabling continuous monitoring, the initiative enhanced operational efficiency and supported the evolution of reservoir models. The platform's ability to generate consistent, validated outputs laid the foundation for future AI-driven innovations in well performance evaluation and intervention planning (Martinez Santiago, et al. 2025).

Well Performance Evaluation and PTA Interpretation

Pressure Transient Analysis (PTA) was automated to extract key reservoir parameters, including static pressure, skin factor, productivity index, and permeability. These outputs were derived from the analysis of high-frequency data collected by Permanent Downhole Gauges (PDHGs), which enabled continuous monitoring of bottomhole flowing pressure (BHFP) and dynamic reservoir behavior. Figure 1 The figure illustrates key well and reservoir parameters, including static pressure, skin factor, and permeability, extracted from PTA interpretation. This data is sourced from the Kappa Automate Pilot-Tepa workshop conducted by TotalEnergies.

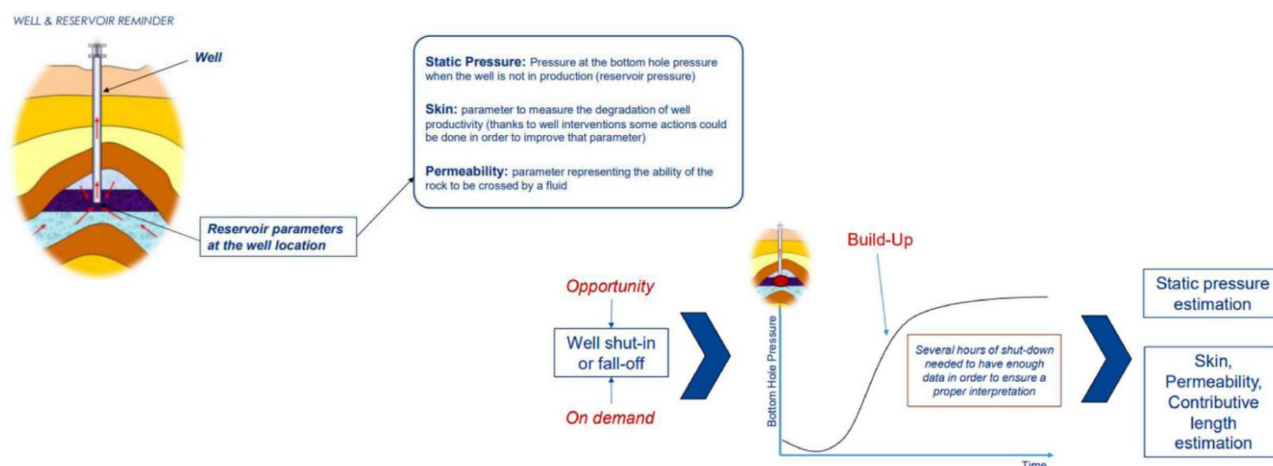


Figure 1—The figure illustrates key well and reservoir parameters, including static pressure, skin factor, and permeability, extracted from PTA interpretation. This data is sourced from the Kappa Automate Pilot-Tepa workshop conducted by TotalEnergies.

The automation of PTA workflows provided timely insights into well performance, allowing engineers to identify flow regimes, assess reservoir boundaries, and detect anomalies such as pressure drops or buildup trends. These insights informed decisions on stimulation strategies, production optimization, and regulatory compliance. For example, wells exhibiting declining productivity indices were prioritized for intervention, with acid stimulation considered in cases where elevated skin values were identified.

The system also supported proactive reservoir monitoring by integrating PTA outputs into the reservoir monitoring plan (RMP). Shut-in durations were optimized based on consistent pressure responses, reducing operational downtime and improving data quality. The ability to track pressure trends over time enabled engineers to validate injection strategies and ensure compliance with reservoir pressure guidelines.

Overall, the automated PTA framework enhanced decision-making by providing reliable, repeatable interpretations that aligned with field observations and regulatory standards. It also improved monitoring efficiency by reducing the need for manual data handling and enabling real-time diagnostics across a network of wells (Martinez Santiago, et al. 2025).

Real-Time Data Acquisition and Event Identification

The foundation of the automated PTA system was a robust data acquisition framework that leveraged real-time inputs from PDHGs installed across the pilot wells. These gauges provided continuous pressure readings, which were ingested into the cloud platform for processing and analysis. To ensure data integrity, smart filtration algorithms were applied to remove noise while preserving critical signal features.

Event detection was a core component of the workflow. The system automatically identified pressure build-ups (PBUs) and fall-offs (PFOs), which are essential for transient analysis. These events were validated using predefined quality control criteria, including duration, stabilization, and rate change thresholds. Once validated, the events triggered automated PTA workflows that synchronized pressure and production data, prepared analysis files, and executed modeling routines.

The automation of event identification eliminated the need for manual shut-in tracking and reduced the risk of missing critical diagnostic windows. Reservoir Engineers could monitor BHFP trends in real time and receive alerts when significant events occurred, enabling faster response times and more accurate reservoir assessments.

This approach also supported multi-well coordination by allowing simultaneous analysis across assets. The platform's centralized repository provided a unified view of event history, analysis status, and interpretation results, facilitating collaboration among engineering teams and improving operational transparency ([Martinez Santiago, et al. 2025](#)).

Automated Pressure Transient Analysis (PTA)

Once events were detected and validated, the system initiated automated PTA workflows that encompassed data preparation, modeling, and result extraction. Pressure and rate data were synchronized and formatted into analysis-ready files, which were then processed using embedded modeling engines. The platform supported both conventional PTA and Incremental PTA (iPTA), allowing for stabilized pressure estimation and advanced diagnostic capabilities.

Model parameters—including reservoir pressure, skin factor, and permeability —were automatically updated based on the latest data inputs. The system applied predefined templates and boundary conditions to ensure consistency across analyses, while also allowing engineers to customize models for specific well conditions. Results were extracted and visualized through interactive dashboards, enabling rapid interpretation and decision-making.

Integration with asset management systems was a key feature of the platform. PTA outputs might be exported to internal databases and linked to other digital tools, such as production forecasting models and intervention planning dashboards. This interoperability ensured that insights from PTA can be incorporated into broader reservoir management workflows and supported cross-functional decision-making.

The automation of PTA not only reduced manual workload but also improved the accuracy and repeatability of interpretations. Engineers could focus on high-value tasks, such as strategy development and anomaly resolution, while the system handled routine analysis and data processing. This shift in workflow dynamics marked a significant advancement in reservoir engineering practices and demonstrated the value of digital transformation in field operations ([Martinez Santiago, et al. 2025](#)).

Results

Reservoir Monitoring and Pressure Support Evaluation

Continuous monitoring of bottomhole flowing (BHFP) and bottomhole close in pressure (BHCP) using Permanent Downhole Gauges (PDHGs) enabled a more granular understanding of reservoir pressure dynamics. The pilot demonstrated that real-time pressure data could be used to evaluate the effectiveness of water injection strategies and their impact on reservoir pressure support. Pressure build-up trends observed during stabilized shut-in periods revealed a direct correlation between injection activity and pressure

stabilization, confirming the role of injection wells in restoring, maintaining or increasing reservoir pressure. **Figure 2**

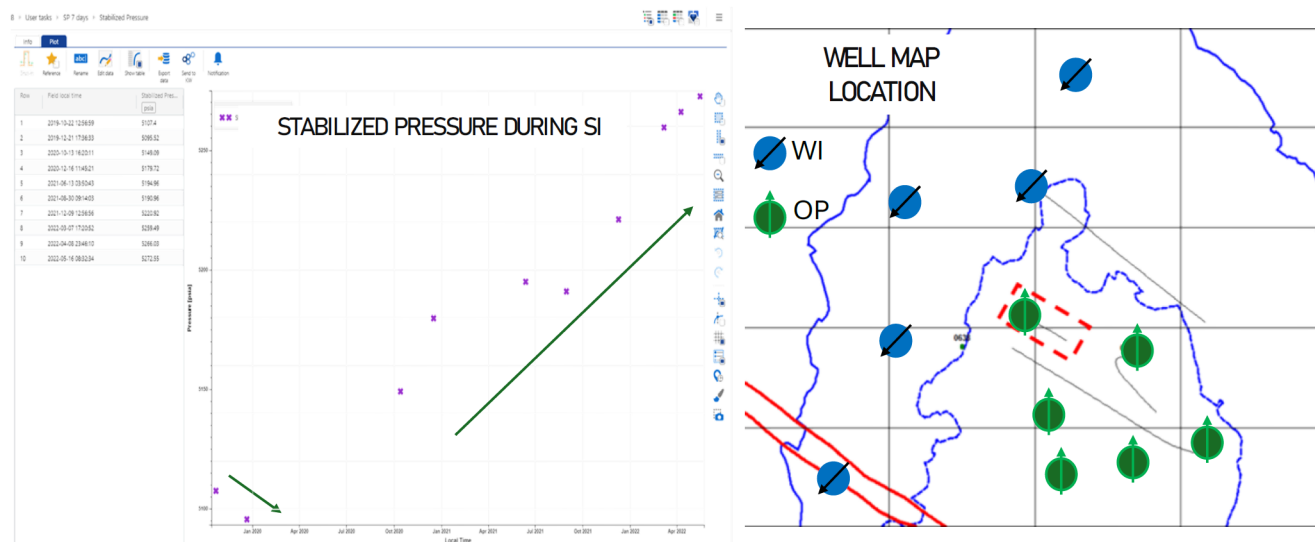


Figure 2—The PDHG installed on the oil producer well highlighted in the red dot rectangle, experienced pressure increments during stabilized shut-in tasks performed, showing a correlation between the water injection incremental rate and the oil producer.

This approach allowed engineers to assess pressure support adequacy without relying solely on periodic well tests. The integration of PDHG data into automated workflows facilitated the identification of pressure anomalies and supported timely adjustments to injection rates. These insights were critical for maintaining reservoir integrity and optimizing production strategies. The ability to track pressure evolution continuously also enabled early detection of reservoir compartmentalization and flow barriers, which are often missed in traditional monitoring setups (Martinez Santiago, et al. 2025)

Well Compliance and Operational Management

The pilot introduced automated compliance monitoring using real-time BHFP data and predefined reservoir pressure thresholds. Alarms were configured to trigger when pressure dropped below acceptable limits, ensuring that wells operated within safe and efficient boundaries. In one instance, a well exceeded its offtake rate, causing BHFP to fall below the compliance threshold. The system flagged the deviation, prompting corrective actions such as rate reduction and choke adjustments. **Figure 3**

This proactive approach to well management reduced the risk of over drawdown and reservoir damage. Engineers were able to respond immediately to deviations, maintaining operational integrity and aligning production with reservoir management guidelines. The automation of compliance checks also reduced the reliance on manual surveillance, freeing up engineering resources for higher-value tasks.

Moreover, the system's ability to log and visualize compliance events provided a historical record for audit and performance review. This feature supported continuous improvement in operational practices and reinforced adherence to reservoir stewardship principles (Martinez Santiago, et al. 2025).

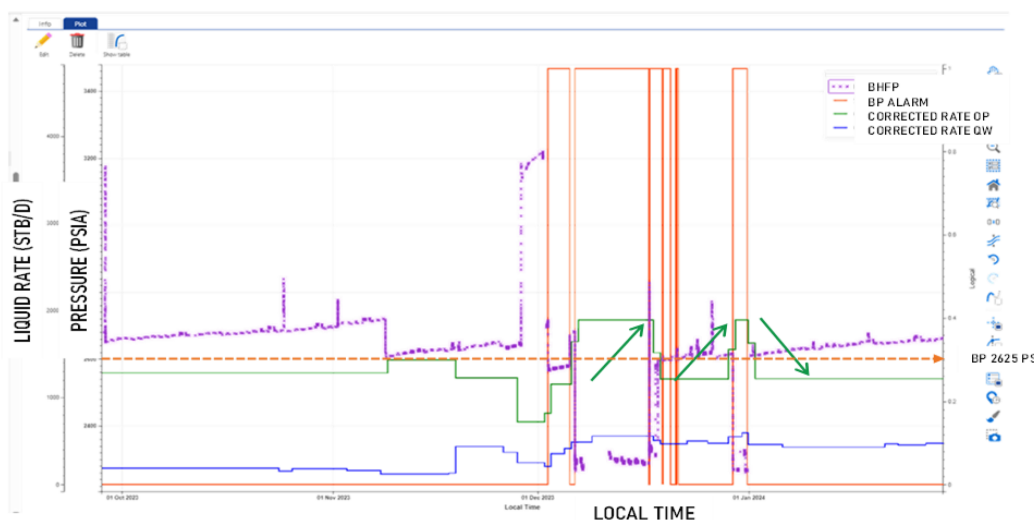


Figure 3—An alarm was created to ensure the well is produced in compliance with Reservoir Guidelines, maintaining the BHFP above 2625 PSI. The alarm was raised in December 2023 during a high offtake period. (red Line)

Reservoir Monitoring Plan (RMP) Optimization

One of the key challenges in reservoir surveillance is determining the optimal shut-in duration for acquiring valid static pressure data. Traditionally, shut-ins of 7 days or longer were used to ensure pressure stabilization. However, the pilot evaluated the feasibility of shorter shut-in periods by comparing pressure responses from 3-day and 7-day durations.

The analysis revealed that for several wells, static pressure values obtained after 3 days were consistent with those from longer shut-ins. This finding enabled a reassessment of the reservoir monitoring plan (RMP), allowing for reduced shut-in durations without compromising data quality. As a result, well availability increased, and production efficiency improved. Figure 4



Figure 4—For some wells, the shut-in period can be reduced, as reservoir pressure calculations showed consistent results for both 3-day and 7-day shut-in durations, enhancing the operational efficiency of the RMP data acquisition process. The ability to model and validate pressure stabilization using PDHG data and automated workflows provided a data-driven basis for optimizing surveillance schedules. This not only enhanced operational flexibility but also minimized production losses associated with extended shut-ins. The pilot demonstrated that digital tools could be leveraged to refine RMP strategies, balancing data fidelity with operational efficiency (Martinez Santiago, et al. 2025).

Well Properties and Stimulation Planning

Automated Pressure Transient Analysis (PTA) enabled the estimation of key reservoir parameters such as skin factor, permeability, and productivity index. These metrics were derived from real-time pressure data

and integrated into stimulation planning workflows. Wells exhibiting high skin values were identified as candidates for acid stimulation, while those with declining productivity indices were flagged for further investigation.

The system also supported temporal tracking of well properties, allowing engineers to monitor changes in reservoir behavior over time. This capability was particularly useful for evaluating the effectiveness of previous interventions and planning future stimulation activities. By consolidating PTA results into a centralized repository, the platform provided a comprehensive view of well performance evolution. [Figure 5](#)

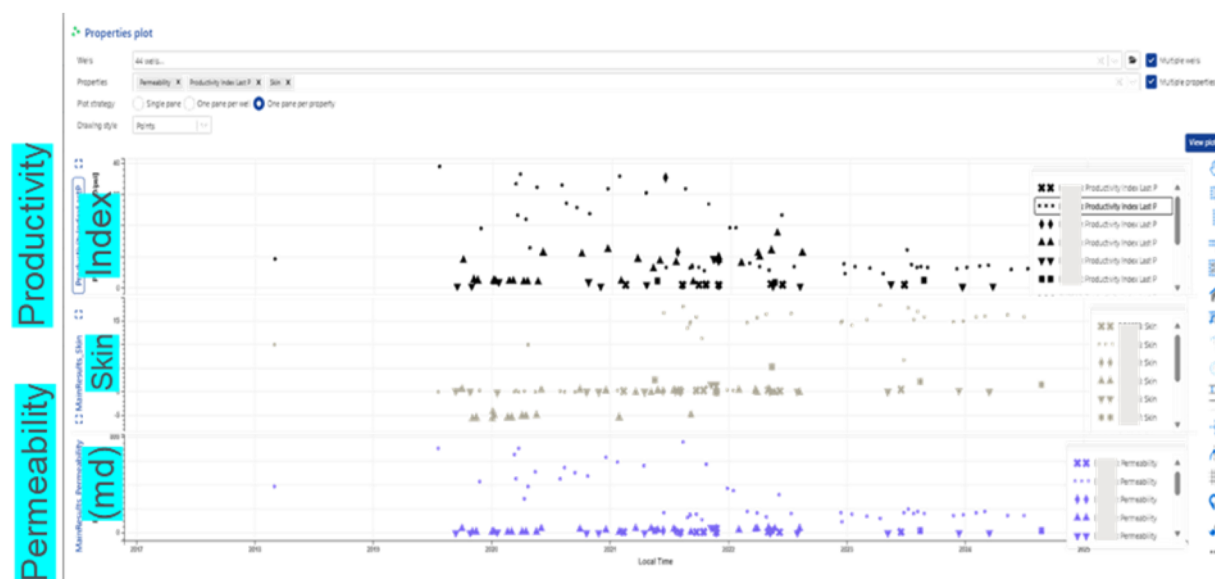


Figure 5—Estimate the 5 wells skin, Permeability and Productivity Index and track evolution over time, then generate a more efficient stimulation plan based on wells or reservoir behavior.

The automation of PTA interpretation reduced analysis time and improved consistency across wells. Engineers could quickly access validated results, enabling faster decision-making and more targeted stimulation strategies. This approach enhanced reservoir recovery and supported long-term field development planning ([Martinez Santiago, et al. 2025](#))

Data Integration and Export

A critical component of the pilot was the seamless integration of PTA results into existing data management systems. The platform was configured to export analysis outputs in formats compatible with internal databases, ensuring that insights were readily accessible to cross-functional teams. Real-time data export enabled efficient reporting and facilitated collaboration between reservoir engineers, production teams, and data management.

The integration of PTA outputs with other digital platforms supported holistic decision-making. For example, pressure trends and well performance metrics can be linked to production forecasting models and intervention planning dashboards. This interoperability ensured that reservoir insights were not siloed but actively contributed to broader operational strategies.

The pilot also demonstrated the value of centralized data repositories for managing PTA documents and results. Engineers could access historical analyses, compare performance across wells, and identify trends that informed future development plans. This level of data integration enhanced transparency, reduced duplication of effort, and supported continuous improvement in reservoir management practices ([Martinez Santiago, et al. 2025](#)).

Discussion

The automated PTA system demonstrated significant advantages over traditional methods. Manual analysis time was reduced by over 50%, and smart filtration decreased data volume by 70% while preserving analytical integrity. The platform enabled faster identification of intervention needs and improved reservoir understanding.

Operationally, the system enhanced compliance, reduced delays, and empowered engineers to focus on high-value tasks. Economically, the pilot showed potential for substantial cost savings and improved resource utilization. Technically, the integration of PDHG data and cloud-based automation marked a leap forward in reservoir management.

Conclusions

The following key insights and achievements summarize the technical and operational value delivered:

- Automation of PTA workflows significantly enhanced operational efficiency, reducing manual analysis time by over 50% and enabling engineers to focus on high-value tasks such as reservoir diagnostics and intervention planning.
- Real-time PDHG data integration enabled continuous reservoir monitoring, allowing for early detection of pressure anomalies, improved injection strategy evaluation, and more responsive reservoir management.
- Compliance and well integrity were proactively managed through automated alarms and pressure threshold monitoring, ensuring wells operated within safe and optimal parameters.
- Reservoir Monitoring Plan (RMP) efficiency was improved by validating shorter shut-in durations, which increased well availability and reduced production downtime without compromising data quality.
- Seamless data integration and export capabilities supported cross-functional decision-making, with PTA results feeding directly into asset management systems, production forecasting tools, and stimulation planning workflows.
- Establishing a strong technical partnership was essential to success, as collaboration with a capable service provider enabled the co-development of customized workflows, ensured secure system integration, and accelerated the adoption of digital tools across multidisciplinary teams.

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The joint efforts between ADNOC Onshore, TotalEnergies Technology HUB, and KAPPA exemplify the value of strategic partnerships in advancing data-driven engineering practices and operational excellence.

Nomenclature

- PDHG – Permanent Downhole Gauge.
- PTA – Pressure Transient Analysis.
- iPTA – Incremental Pressure Transient Analysis.
- BHFP – Bottomhole Flowing Pressure.

- PBU – Pressure Build-Up.
- PFO – Pressure Fall-Off.
- RMP – Reservoir Monitoring Plan.
- Skin Factor – A dimensionless parameter indicating formation damage or stimulation near the wellbore.
- Permeability – A measure of the reservoir rock's ability to transmit fluids.
- Productivity – A ratio of production rate to pressure drawdown, used to assess well performance.
- Index
 - AI – Artificial Intelligence.
 - QC – Quality Control.

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